SOC Log Monitoring & Threat Detection Using Splunk

Project Type: Hands-on Cybersecurity / SOC Lab Project

Platform: Ubuntu Linux

SIEM Tool: Splunk Enterprise

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1. Project Introduction

The objective of this project is to understand how Security Operations Center (SOC) analysts monitor system logs and detect suspicious activities using a SIEM tool.

In this project, Splunk Enterprise was configured on Ubuntu Linux to collect and monitor authentication logs. Failed login attempts were detected, visualized through a dashboard, and monitored using an alert.

2. Tools and Technologies Used

- Ubuntu Linux
- Splunk Enterprise
- Oracle VirtualBox
- Linux Authentication Logs ("/var/log/auth.log")
- SPL (Search Processing Language)

3. Log Collection and Configuration

Splunk was configured to monitor the Ubuntu authentication log file:

"/var/log/auth.log"

The log input was added successfully with the following configuration:

- **Index:** main

- **Source Type**: Ubuntu_Auth_Logs

- **Host:** ubuntu

After configuring the input, authentication events were successfully collected in Splunk.

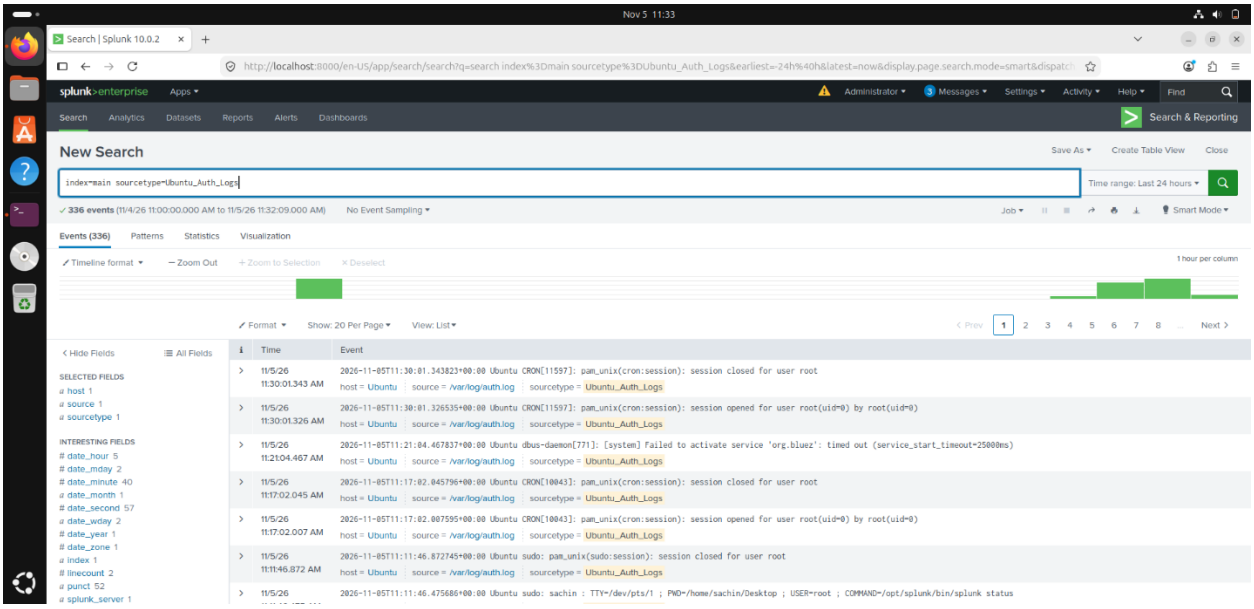


Figure 1: Ubuntu Authentication Logs

4. Failed Login Detection

A failed login attempt was generated in Ubuntu by entering an incorrect password.

The following SPL search was used to identify authentication failures:

index=main sourcetype=Ubuntu_Auth_Logs "authentication failure"

Splunk successfully identified failed authentication events.

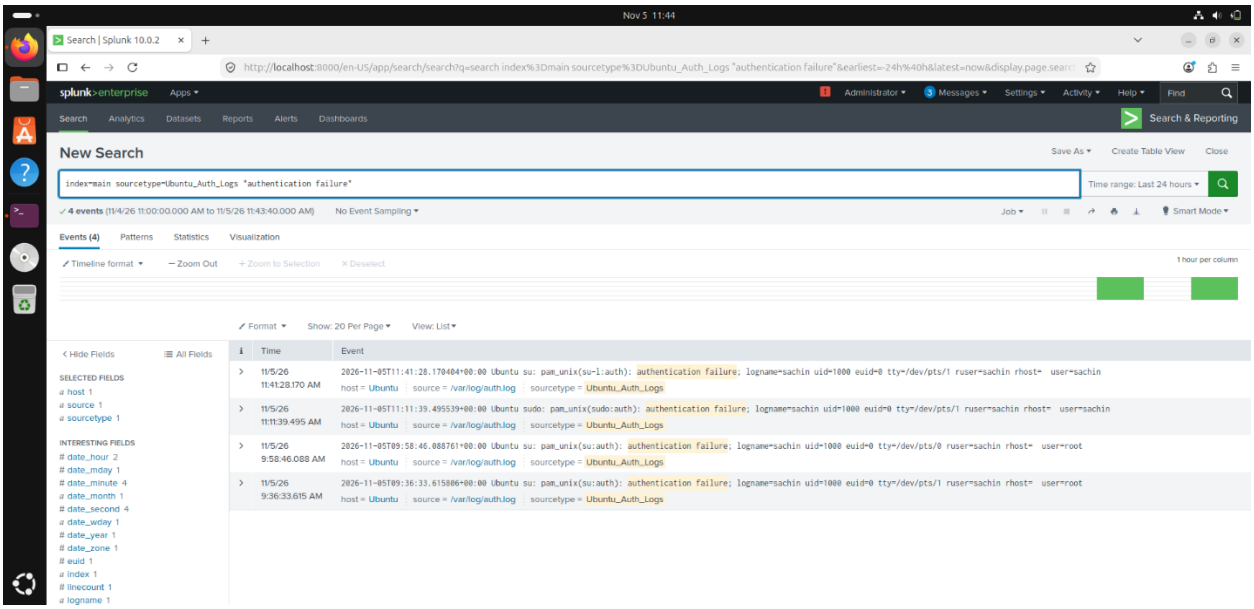


Figure 2: Authentication Failure Detection

5. Authentication Failure Timechart

A timechart was created to visualize authentication failure events over time.

index=main sourcetype=Ubuntu_Auth_Logs "authentication failure"

| timechart count

This visualization helps understand when authentication failures occur.

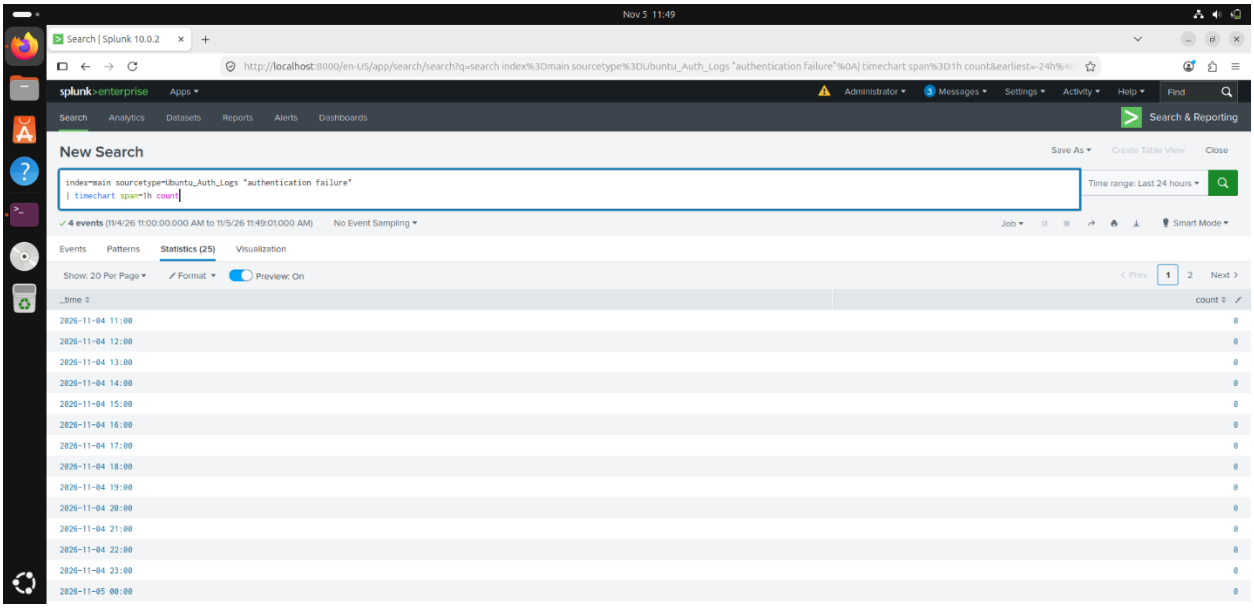


Figure 3: Authentication Failure Timechart

6. SOC Monitoring Dashboard

A Splunk dashboard named SOC Monitoring Dashboard was created.

It contains two panels:

1. Authentication Failure Monitoring

2. Total Log Events

The dashboard provides a simple overview of collected authentication events and failed login activity.

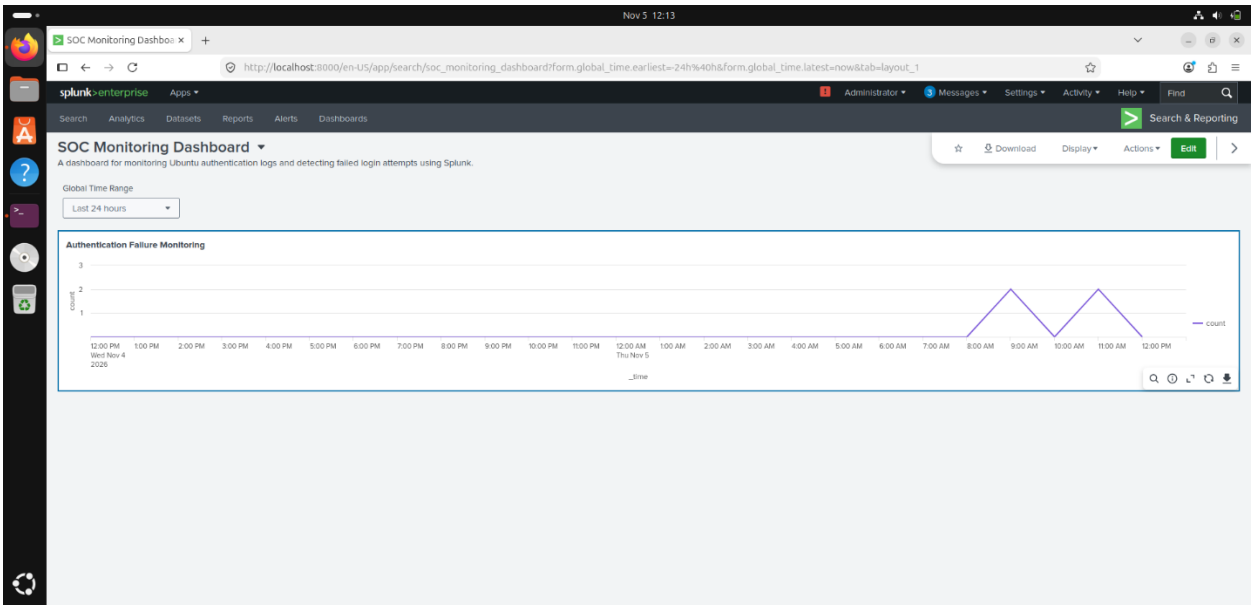


Figure 4: SOC Monitoring Dashboard

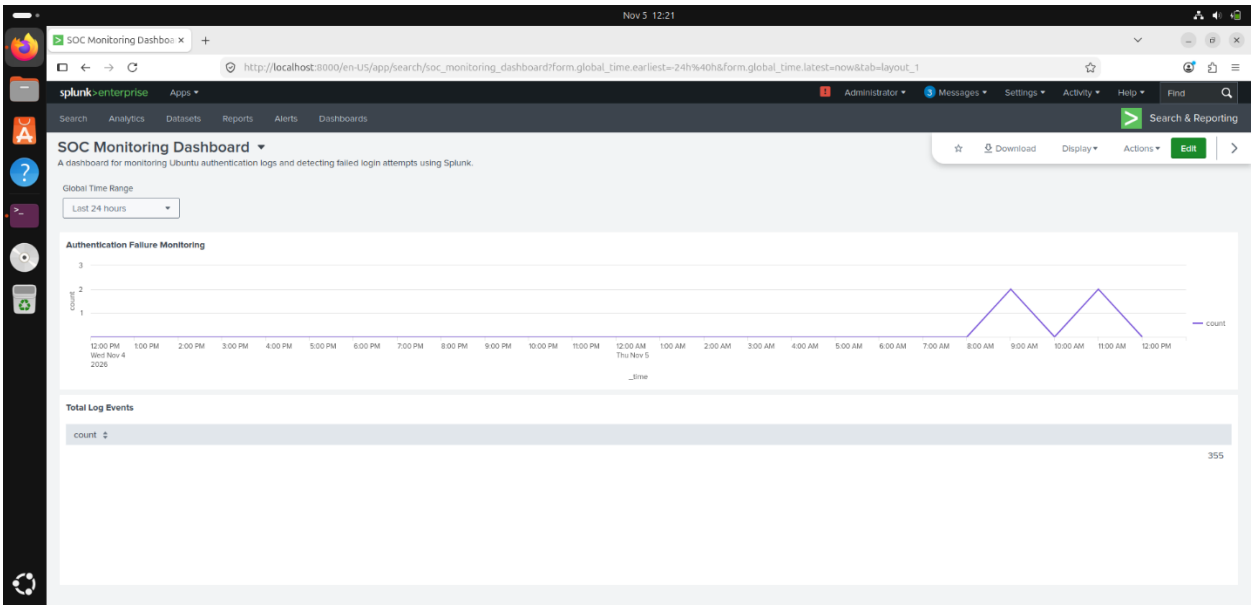


Figure 5: Total Log Events

7. Alert Configuration

An alert named Ubuntu Failed Login Detection was configured in Splunk.

Alert Description: Detects failed authentication attempts on Ubuntu using Splunk logs.

The alert was configured to monitor failed authentication events and add triggered alerts to Splunk's Triggered Alerts section.

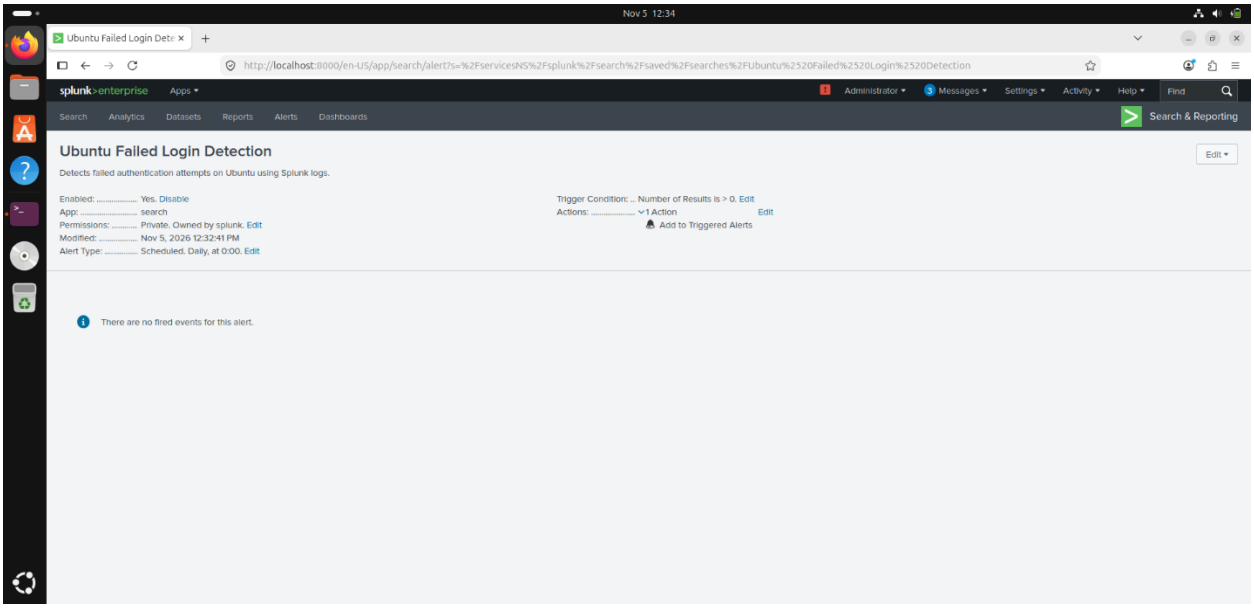


Figure 6: Ubuntu Failed Login Alert Configuration

8. Alert Testing and Verification

To test the alert, an incorrect password was entered during a Linux authentication attempt.

The authentication failure was recorded in Ubuntu logs and detected by Splunk.

The alert appeared in the Triggered Alerts section, confirming that the detection workflow was working during the test.

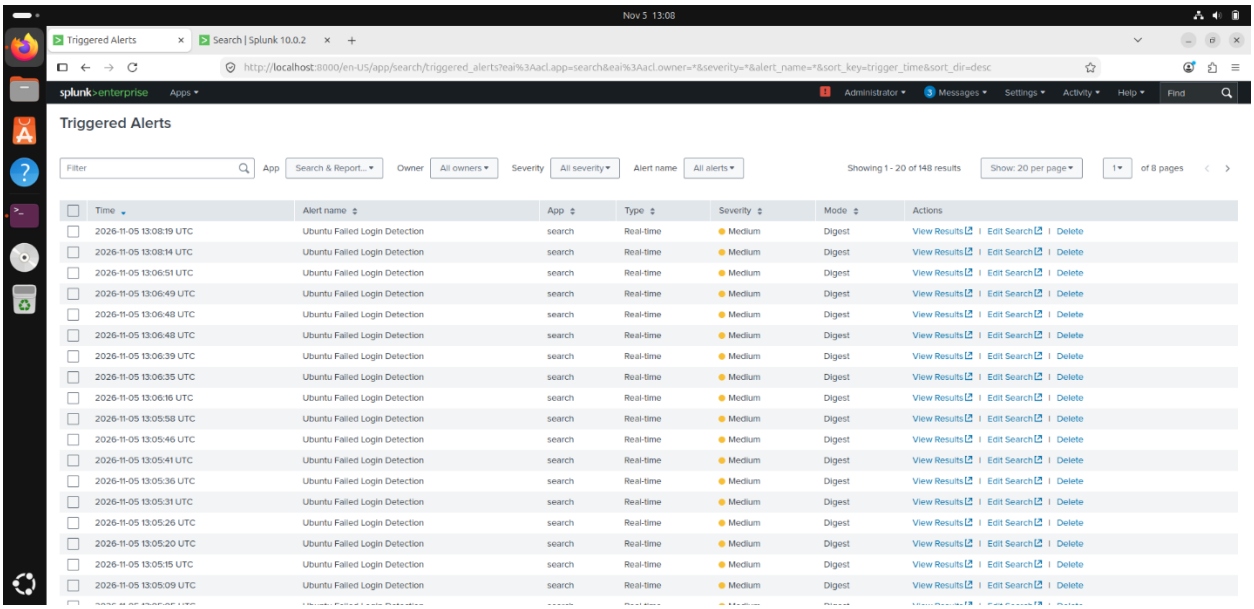


Figure 7: Splunk Triggered Alerts List

9. Learning Outcomes

Through this project, I learned:

- How to collect Linux authentication logs in Splunk.
- How to search and analyze security events using SPL.
- How to identify failed login attempts.
- How to create a monitoring dashboard.
- How to configure and test alerts.
- Basic SOC monitoring and log analysis concepts.

10. Conclusion

This hands-on project helped me understand the basic workflow of a Security Operations Center (SOC).

I gained practical experience in log collection, security event monitoring, failed login detection, dashboard creation, and alert configuration using Splunk Enterprise.

This project strengthened my understanding of SIEM tools and security monitoring.

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